Application of Machine Learning in Solving Dynamic Flow Problems

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Abstract

The lid-driven cavity problem is a classic benchmark problem in computational fluid dynamics domain that involves simulating the flow within a square cavity with a moving lid. The present study addresses the lid-driven cavity problem using two different approaches: Physics-Informed Neural Networks (PINNs) and the Finite Difference Method (FDM). The main goal is to investigate and compare how well these methods work to simulate fluid flow inside a cavity driven by a lid. The Navier-Stokes equations are discretized using the Finite Difference Method and are then solved numerically on a staggered grid, providing a baseline numerical solution for the fluid flow within the cavity. In parallel, without using grid discretization, a Physics-Informed Neural Network (PINN), is employed to approximate the underlying fluid dynamics. A Feed Forward Neural Network is used in this study to approximate the flow field in this problem. A thorough investigation of the PINN approach is conducted in this paper.

Keywords : Lid Driven Cavity Flow, Physics Informed Neural Network (PINN), Finite Difference Method (FDM)

1 Introduction

Lid-driven cavity flow is a benchmark problem in the field of computational fluid dynamics, with its applications in vortex-induced vibrations. Hence, reasonably accurate and efficient simulation techniques are necessary. The traditional approach includes the Finite Difference Method, which works by discretizing the Partial Differential Equations (PDE) over a staggered grid system. This can result in an increased cost of computation where the Incompressible Navier-Stokes equations are solved over a staggered grid.

The recent advancements in Machine Learning and the increasing experimentation of various data-driven techniques in the domain of scientific simulations have opened up avenues that were unknown 10-20 years ago. A lot of these developments in the application of Machine Learning over various domains are often heavily dependent on high-quality data. When addressing such physical systems with Machine Learning, the fact that we already know some prior physics constraints often gets neglected. This prior knowledge about the physics of the system overwhelmingly aids the training process and can even produce results with little or no actual system data at all. Such networks form a huge class of data-driven modelling techniques called Physics-Informed Machine Learning (PIML).

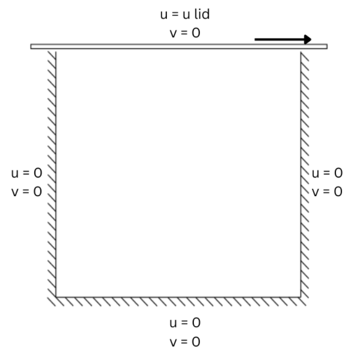
The very pioneering work in this domain involved solving nonlinear partial differential equations (PDEs) (2019) [1], which used existing knowledge of physics to solve nonlinear partial differential equations. Many such works followed in the domain of applications of Machine Learning Methods to solve differential equations. One such work given in [2] (2022) was replicated, as a lead-up to this work, where higher-order and highly non-linear PDEs were solved using Physics-Informed Neural Networks (PINN).A study conducted on PINNs in [5] (2024) covered an in-depth exploration of PINNs in general followed by its future prospect. The paper served as one of the very first motivators for exploring and diving deep into the PINNs and its applications. The increasing hardware capacity and the ever-increasing source of data can be largely attributed to the developments of such data-driven methods. A high-level application of PINNs is demonstrated in a Fluid Structure Interaction (FSI) problem in [3] (2024), where a very intricate study was conducted on the hyperparameters of the formulated PINN. This served as another motivation for the present work, which was drawn primarily to analyze the fine hyperparameters of a formulated PIML model, as there is no defined set of hyperparameters or any unified theory for it, and it is highly dependent on the problem that is being addressed through the PINN. The application space of PINNs can also be well and truly extended to turbulent flows and modelling, especially in areas of flow reconstruction as described in [4] (2024), owing to the ability of PINNs to learn very sparse data with a very reasonable accuracy due to its knowledge of the underlying physics of the system.

In the present study, an attempt is made to develop and apply a PINN to solve the steady incompressible Navier-Stokes equation for lid-driven cavity flow. This problem was chosen apart from being a benchmark problem in Computational Fluid Dynamics (CFD) because it offers significant complexities and constraints to account for in its result and training process.

Subsequent sections provide a detailed account of the problem setup, results, and observations.

2 Problem Definition

The lid-driven cavity flow is a simple problem setup where a fluid flows in a square cavity driven by a top lid. The lid shears the upper layer of the fluid, which sets the flow in motion, resulting in a vortex shedding in the fluid in the cavity. A distinct boundary layer effect is also observed as the top layer of the fluid sets in motion with the same velocity as the lid. A diagrammatic representation of the problem is given in figure (**??**)



- Leftward velocity at the upper boundary to maintain relative velocity between the solid and fluid surface as zero.
- No-slip condition at rest 3 walls. Horizontal (u) and vertical (v) components of the velocities are 0.

The studied system is governed by the 2-D incompressible steady Navier-Stokes equation given by

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{1}{\phi}\frac{\partial p}{\partial x} - \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 0 \quad (x - Direction) \tag{1}$$

$$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{1}{\phi}\frac{\partial p}{\partial y} - \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) = 0 \quad (y - Direction) \tag{2}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (Continuity\ \ Equation) \tag{3}$$

where equations (1) and (2) are the momentum equations in x and y directions, respectively, and (3) is the continuity equation.

3 Physics Informed Neural Network (PINN) for Navier Stokes Equations

A Feed Forward Neural Network (FNN) is used for the problem. The network receives x and y as input and gives the stream function ψ, which is a function of u and v, and the pressure field p as output. The input data to the neural network is divided into two parts: the boundary data and the bulk data within the cavity. The boundary data consists of $10,000$ randomly sampled x and y points at the boundary and their corresponding u, v, and p values as the output, which are previously known from the no-slip condition. This is used to calculate the two components of the loss function (discussed later). The bulk data is taken inside the cavity, and $10,000$ random points (x, y) are sampled within the cavity. The network predicts u, v, and p for each of those points. This output is then substituted into the Navier-Stokes equations to find the PDE residue. The derivative terms of the Navier-Stokes equations are calculated using TensorFlow's tf.GradientTape() function.

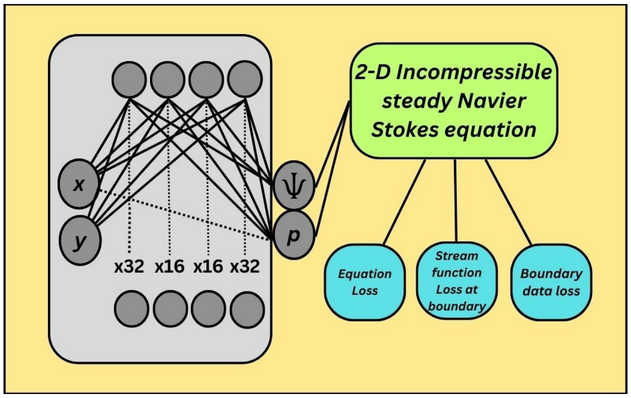


Figure 2 : Schematics of the (PINN) used

3.1 Loss formulation

In the loss formulation, a typical mean squared error loss (MSE loss) will be used to calculate the loss. The general form is given by

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m} ||f(x_i) - y_i||^2 \tag{4}$$

where J is the loss function, w is the weight and b is the bias for the function $f(x)$ having m data points.

In this problem, the loss function is designed as the weighted sum of the 3 loss components, namely L_{eq}, L_ψ, and L_{bnd} as given in equation (5).

$$L = \alpha L_{eq} + \beta L_\psi + \gamma L_{bnd} \tag{5}$$

where L is the total loss and L_{eq} , L_ψ and L_{bnd} are equation loss, stream function loss and boundary data loss respectively and α, β and γ being their weights respectively.

The L_{eq} for x and y momentum equation residuals is expressed as L_{eq_x} and L_{eq_y} for N number of points within the cavity and is given by

$$L_{eq_x} = \frac{1}{2N}\sum_{i=1}^{N} ||r_m^x(x_{body}^i)||^2 \tag{6}$$

$$L_{eq_y} = \frac{1}{2N}\sum_{i=1}^{N} ||r_m^y(x_{body}^i)||^2 \tag{7}$$

where x_{body} signifies that the equation loss is calculated on the sampled points within the cavity domain i.e, the bulk data loss. The r_m^x and r_m^y are the PDE residue for the momentum equation and are given by

$$r_m^x(x_{body}^i) = uu_x + vu_y - p_x - \frac{1}{Re}(u_{xx} + u_{yy}) \tag{8}$$

$$r_m^y(x_{body}^i) = uv_x + vv_y - p_y - \frac{1}{Re}(v_{xx} + v_{yy}) \tag{9}$$

The L_ψ is the loss of the stream function at the boundary points, which is already a known value for the problem studied. The stream function is given by

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x} \tag{10}$$

where u and v are the velocities in x and y direction respectively. Additionally the continuity equation in terms of stream function satisfies

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{11}$$

Let the equation residue for the continuity equation be called r_c.

$$L_\psi = \frac{1}{2N} \sum_{i=1}^{N} ||r_c^y(x_{body}^i)||^2 \tag{12}$$

where,

$$r_c^y(x_{body}^i) = u_x + v_y \tag{13}$$

Lastly, for the data loss, L_{bnd} is given by

$$L_{bnd} = \frac{1}{2N} \sum_{i=1}^{N} ||f(x_{bnd}^i, y_{bnd}^i) - \hat{y}||^2 \tag{14}$$

where, $\hat{y}$ is the true value of u and v i.e, the x and y velocity at the boundary points.

At different values of α, β and γ the model is trained and outputs are obtained and hence the impact of the weights of the different loss components on the training curve and the results are analyzed.

3.2 Optimization

In the training of the PINN one of the most crucial aspects is the choice of optimizer used. The choice of the optimizer often depends on the formulation of the PINN and also on the problem that is being investigated. In this case, like most other PINNs the L-BFGS (Limited memory Broyden–Fletcher –Goldfarb–Shanno) optimizer is used. This is primarily because of the fact that a BFGS optimization is very superior to other Stochastic Gradient Descent based optimizers in navigating complex PDE loss landscapes and its ability to converge in very less iterations. Cases involving complex PDEs like Navier Stokes equations often result in a very non convex and stiff curves for the loss functions, which leads to oscillations and high convergence time for optimizers like Adam.

3.3 Final Training Configuration

A final configuration of the set up is provided with all the hyperparameter settings before initiating the training process.

- **Problem configuration:** Lid velocity (u_{lid}) = 1 and Re = 100
- **Network Architecture:** 2 node input followed by 32, 16, 16, 32 hidden layers and 2 node output layer

- **Optimizer:** L-BFGS
- **Curvature History:** m = 50
- **Learning Rate:** maxls = 50
- **Iterations:** 5000
- **Number of Training Data:** 10000
- **Number of Test Data:** 100
- **Activation:** swish

4 Results

4.1 Analysis of the PINN performance with respect to Finite Difference Method (FDM)

The lid-driven cavity problem for $u_{lid} = 1$ and $Re = 100$ is first solved using the finite difference method (FDM) in a staggered grid system. Grid size is so selected that it does not hamper numerical convergence. The results of the FDM solution are shown in Figure 3. The results for the same problem solved with PINN are shown in Figure 4. Figures 5(a) and 5(b) are the streamlines, and 5(c) is the convergence plot during the training process of the PINN.

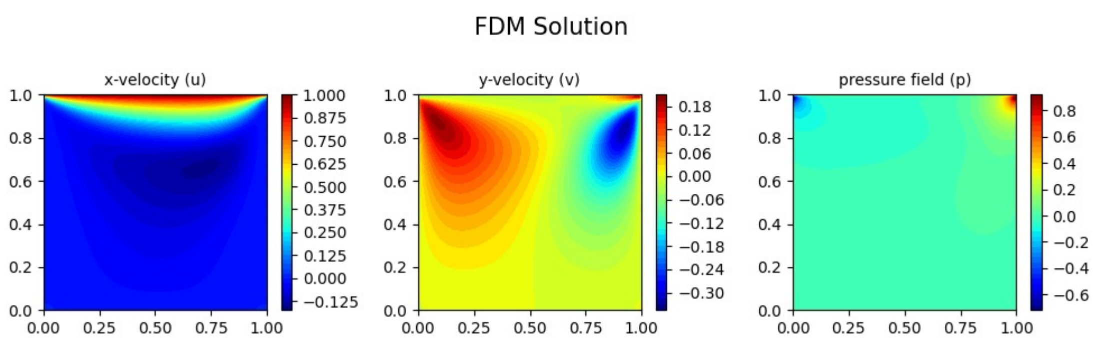


Figure 3

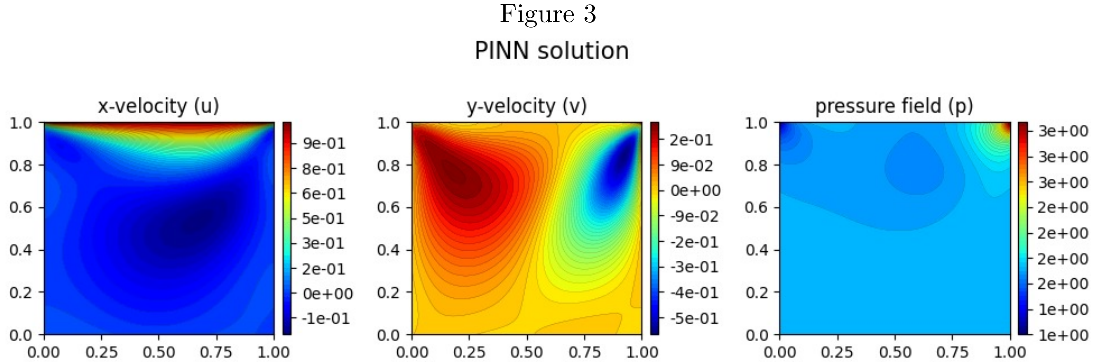


Figure 4

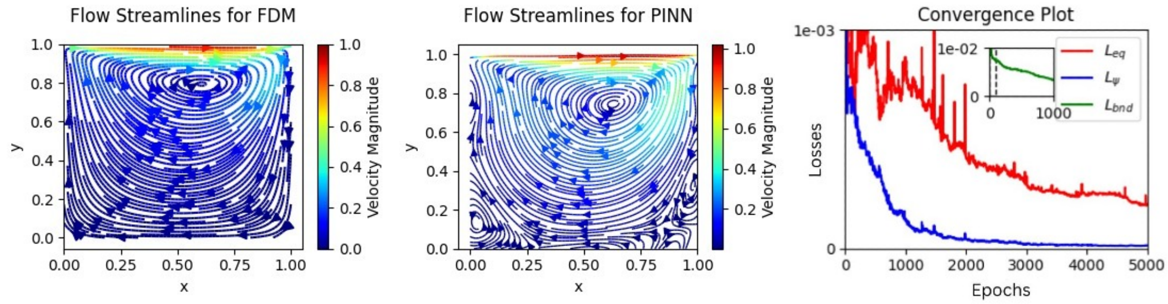


Figure 5(a), 5(b) and 5(c) from left to right

The streamline plot from the PINN is found to be in very good agreement with the FDM solution. The FDM solution was simulated using a very coarse grid with the sole objective of providing a reference for the evaluation of the PINN solution. This resulted in the FDM's inability to capture the secondary vortices in the cavity. Surprisingly, the PINN, on the other hand, managed to capture the secondary vortex fluctuations in the cavity with very reasonable accuracy.

A quantitative study on the impact of weights of the loss components is provided in the following subsection.

4.2 Impact of Loss weighting (α, β and γ) on model accuracy and convergence

From Equation (5) we get, the total loss is the weighted summation of the equation loss (L_{eq}), stream function loss at boundary points L_{ψ}, and boundary data loss L_{bnd}. To understand the network's sensitivity to each losses a manual tuning of the weights α, β and γ have been carried out.

Serial No.	α	β	γ	L_{eq}	L_{ψ}	L_{bnd}	L
1	1.0	1.0	1.0	0.00022	1.39e-5	0.0017	0.002
2	1.0	1.0	0.01	1.33e-5	1.15e-5	5.5e-5	7.9e-5
3	1.0	1.0	0.1	6.9e-5	2.7e-5	0.00034	0.0004
4	0.01	0.01	1.0	0.0001	1.03e-7	0.0068	0.006801
5	0.1	0.1	1.0	0.00013	7.08e-7	0.00064	0.00078

Table 1: Loss weighting sensitivity of the PINN

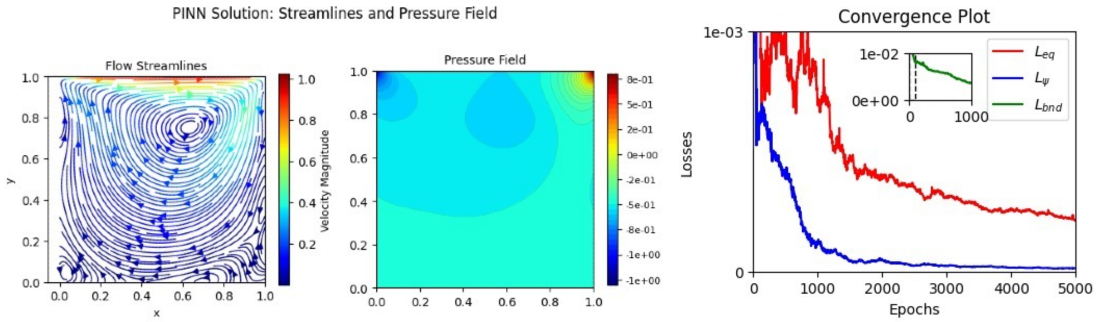


Figure 6: Results for $\alpha = 1$, $\beta = 1$ and $\gamma = 1$ with streamline, pressure field, and convergence from left to right

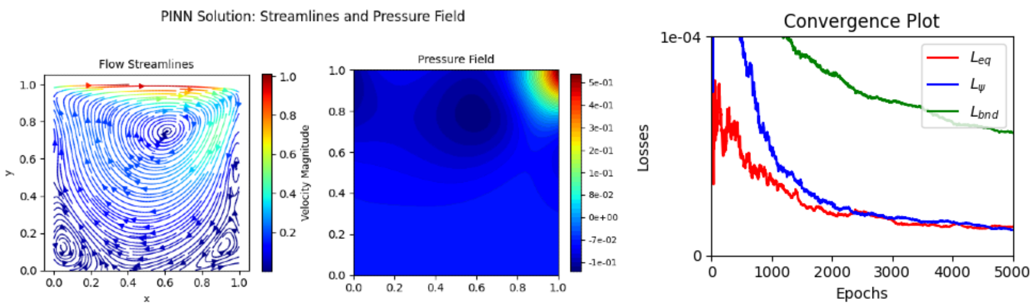


Figure 7: Results for $\alpha = 1$, $\beta = 1$ and $\gamma = 0.01$ with streamline, pressure field, and convergence from left to right

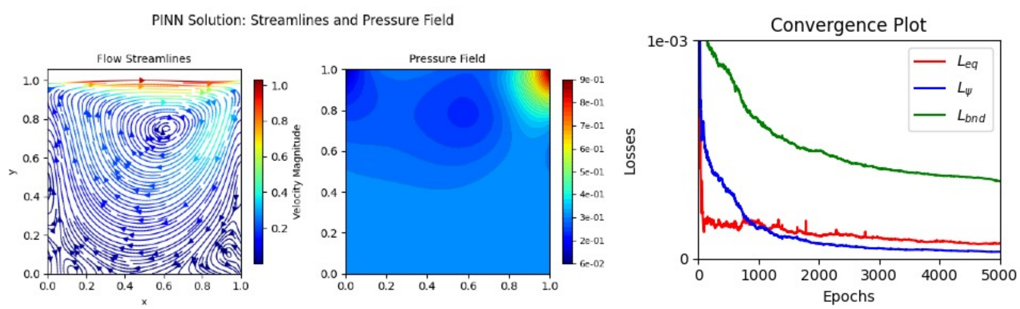


Figure 8: Results for $\alpha = 1$, $\beta = 1$ and $\gamma = 0.1$ with streamline, pressure field, and convergence from left to right

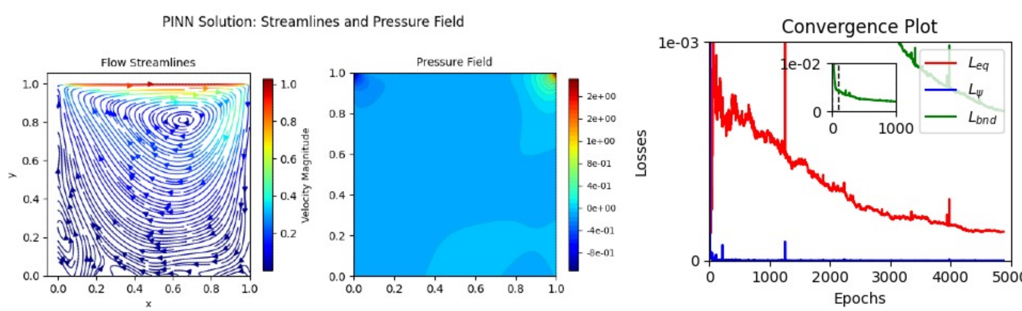


Figure 9: Results for $\alpha = 0.01$, $\beta = 0.01$ and $\gamma = 1$ with streamline, pressure field, and convergence from left to right

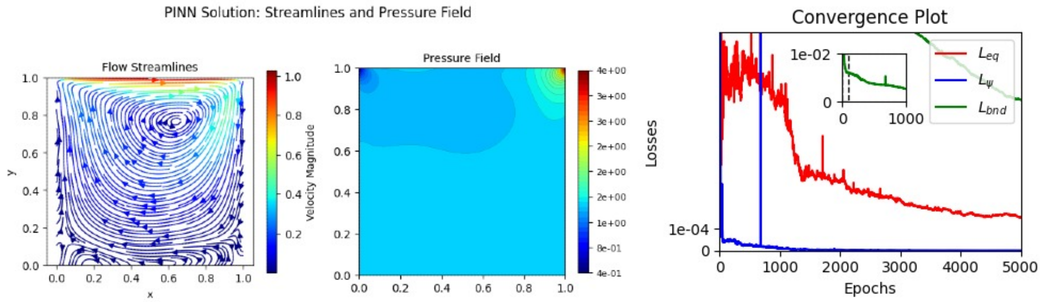


Figure 10: Results for $\alpha = 0.1$, $\beta = 0.1$ and $\gamma = 1$ with streamline, pressure field, and convergence from left to right

The results above show the sensitivity of the model's accuracy and convergence for different values of α, β, and γ. In Figure 7, when $\alpha = \beta = \gamma = 1$, the model results in fairly accurate results when compared to the coarse grid FDM simulation. It has to be noted that the pressure field is the most sensitive to the model's hyperparameters and the coefficients of the losses. When the boundary data loss is minimized, i.e., when $\gamma = 0.01$, the model manages to predict the secondary and tertiary vortices. In fact, it is clearly observed that the vortex phenomena, especially the resolving of the secondary vortex and tertiary vortex, are heavily influenced by γ. In figure 7 and figure 8 we can see when $\gamma = 0.01$ or 0.1 the smaller eddies are resolved and more so in figure 7 where γ is the smallest. This can be explained as, for a fairly simplistic boundary condition, a higher value of gamma will result in the forceful enforcement of the boundary no-slip condition at the near wall locations, often at the expense of resolving near-wall interior flow physics. So a γ value of 0.01, which is one-one-hundredth of the α and β, is always just the perfect enforcement of the boundary condition for fairly simple geometry based problems like cavity flow.

For all the cases we can see the boundary data loss L_{bnd} is significantly higher than the equation loss except for number 4 where $\gamma = 0.01$. Also, it is clearly observed that the model has the best accuracy in terms of loss for $\gamma = 0.01$ and second best accuracy for $\gamma = 0.1$.

5 References

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